

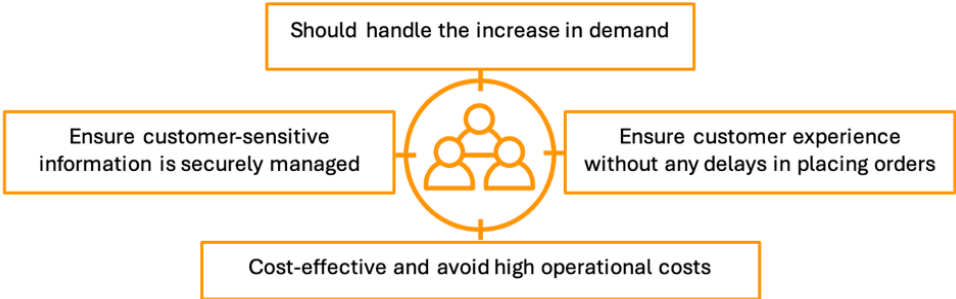
# Design Document

## Solution Objectives

The café’s web application aims for high availability across multiple Availability Zones to ensure uptime during issues. Scalability is crucial for handling traffic spikes, with efficient load balancing to prevent bottlenecks. The solution is also designed for low maintenance, allowing the café’s team to focus on business operations while ensuring data integrity and security compliance. The final goal of the solution is to provide a seamless user experience, allowing customers to interact with the café’s website without delays or interruptions, even during peak traffic periods.

## Business Decisions

What are Sofia and Nikhil’s café website needs?



How can AWS services respond?

- 1. Low Cost (discussed in the Cost Estimate section)
- 2. Security Compliance (discussed under Security Through Bastion Host and WAF)
- 3. Amazon Elastic Compute Cloud (Amazon EC2)

Why EC2? Comparison with the alternative, AWS Elastic Beanstalk.

|                | ✅ AWS EC2 (AWS 2024)                                                                                                                                 | AWS Elastic Beanstalk (AWS 2024)                                                                                                           |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Flexibility    | Provides various instance types and configurations, enabling users to select the best fit for their applications. Users can also create custom AMIs. | Tailored for specific use cases, may not meet all application requirements, requiring developers to adjust their applications accordingly. |
| Vendor Lock-in | Can be ported to other cloud providers more easily.                                                                                                  | Difficult to migrate to other platforms.                                                                                                   |
| Scalability    | Users have finer-grained control.                                                                                                                    | Limited to platform defaults                                                                                                               |
| Cost Savings   | Can be optimised by choosing specific instance types and right-sizing instances based on needs.                                                      | Users pay for the underlying AWS resources, which may lead to higher costs due to resource bundling.                                       |

Sofia and Nikhil expect 10k+ users, which requires scaling resources to handle user fluctuation. **AWS Auto Scaling** monitors applications and adjusts capacity to maintain steady (AWS 2024). It can dynamically increase resources during these spikes and then scale down afterwards whenever there is lower traffic helping to ensure cost optimisation and resource efficiency.

**AWS Load Balancing** was also included to optimise the performance of servers, ensuring that users are spread evenly. This is essential to ensure the website is always up and running, creating a seamless and lag-free user experience for her customers.

## 4. Virtual Private Cloud

Why NAT Gateway? Comparison with the alternative, Virtual Private Network (VPN).

| ✔ NAT gateway (AWS 2024) |                                                                          | VPN (Paloalto 2024)                                                                                    |
|--------------------------|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Ease of Set up           | Simple setup and integration with private subnets in a VPC.              | Complex setup requiring VPN configuration on both AWS and on-premises (or external) networks.          |
| Scalability              | No manual intervention needed.                                           | Struggle to scale smoothly due to their dependence on hardware and static configurations               |
| Cost Savings             | Predictable cost structure with per-hour and per-GB data processing fee. | Not cost-effective, can incur higher expenses compared to the adoption of cloud-native SASE solutions. |

Having multiple availability zones ensures that AWS and the website remain operational even if one zone has issues. The **NAT gateway** allows instances without public addresses to access the internet, keeping the website accessible during high-traffic times.

### Services adopted

Due to the limitations on accessibility, our current solution currently only integrates NAT Gateways, EC2 Auto Scaling and Elastic Load Balancing.

### NAT Gateways

The implementation of a NAT gateway addresses the “Anticipate Failure” principle of Operational Excellence. By developing NAT gateways across 2 Availability Zones, traffic can be rerouted through another gateway if one fails, minimising downtime and disruption, and enhancing reliability. The deployment of NAT gateways also adheres to the “Mechanical Sympathy” principle of Performance Efficiency. NAT gateways handle high levels of network traffic, making them ideal for directing outbound internet traffic from instances in private subnets without exposing to inbound internet threats.

Regarding how deployment risk can be mitigated, Amazon CloudWatch can monitor the NAT gateways performance and traffic metrics, with automated alerts to notify the team of irregularities to allow quick responses (AWS 2024).

In anticipation of catastrophic failure impacting the NAT gateway, contingency plans are also in place, including the addition of NAT gateways in alternative regions, ensuring cross-region replication for critical traffic, and automatic failover to backup NAT gateways in other zones. These measures contribute to the stability and operational continuity of the café’s infrastructure, aligning with our commitment to delivering business value through resilient operations.

### EC2 - Auto Scaling & Elastic Load Balancing

The design principle “Scale Horizontally to Increase Aggregate Workload Availability” of Reliability is enabled by both Auto Scaling and Elastic Load Balancing (ELB). Horizontal scaling involves adding more instances to meet increased demand rather than relying on a single powerful instance. Auto Scaling automatically deploys additional instances during demand spikes, while the ELB evenly distributes traffic to prevent any single instance from becoming overloaded. This design enhances reliability by ensuring smooth operation even if individual instances fail. Moreover, the implementation of Auto Scaling and ELB addresses the “Perform Operations as Code” principle of Operational Excellence. By establishing Auto-Scaling policies as code, we can specify criteria for scaling up and down, ensuring that resources are consistently optimized. ELB configurations are also automated, facilitating efficient deployment and modifications across availability zones, which help provide a smooth experience for customers during high-demand periods.

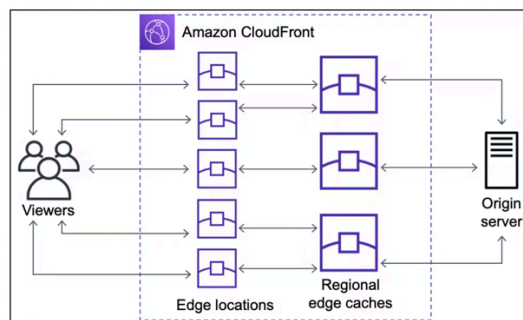
To “design interactions in a distributed system to mitigate/withstand failures”, EC2 Auto Scaling and Elastic Load Balancing are utilised. Auto Scaling adjusts the number of instances based on demand and health checks and can handle sudden load change. This ensures that the website's performance is consistent and avoids single points of failure. To “manage workload and operations events” EC2 Auto Scaling groups are configured across multiple Availability Zones, ensuring that if one zone experiences a failure, instances in another zone can take over. Additionally, regular backups and automated failover mechanisms ensure data integrity and minimize downtime, aligning with our goal of delivering a resilient and reliable experience for the café's customers.

## Services that can be adopted in the future

To further ensure high availability and reduce latency that customers encounter when accessing the website, services like Amazon CloudFront, Relational Database Service (RDS) and CloudWatch can be implemented.

### CloudFront

Given that the café may experience an influx of customers upon the broadcast, implementing a content delivery network through CloudFront is a beneficial layer to add to the existing solution. CloudFront provides reduced latency through global edge locations, where requests are routed directly to the customer's closest edge location when the website is accessed. This significantly accelerates webpage loading as requesting data from the origin server takes longer to respond due to the complex network collection (Rabelo 2023).



Leveraging Amazon CloudFront meets the “Go global in minutes” principle of Performance Efficiency. With CloudFront, the café can instantly deliver content to customers around the world, as it operates with a network of edge locations. This design principle ensures that content reaches users quickly, which aligns with the Performance Efficiency pillar by minimizing latency and maximising availability across different regions. With CloudFront, users benefit from both speed and reliability, ensuring a seamless experience when they access the café's website from wherever they are.

CloudFront was selected as the “best-performing architecture” as it offers a highly efficient architecture for distributing content, directing users to the nearest edge location, improving user experience through faster content delivery.

In anticipation of a catastrophic failure affecting a specific edge location, CloudFront's automatic failover mechanism can reroute traffic to other available edge locations. By enabling detailed monitoring and configuring alerts in CloudWatch, the operations team can proactively address performance issues and maintain consistent service availability.

### RDS

Given that scalability and availability are key goals for Sophia and Nikhil, Amazon Relational Database Services (RDS) is an additional service that can be considered. RDS offers increased website durability and availability through the deployment of a Multi-AZ database instance, which replicates data into a different availability zone for standby in case of a failure in the primary zone. For the café, the RDS has been added under availability zone: us-east-1b as the standby for the primary availability zone us-east-1a. Through Multi-AZs, the reliability parameter is addressed, offering the café a resilient website that has an SLA of up to 99.95% is provided whilst also further reducing latency. This will help guarantee the availability of the website especially in times of high traffic that is anticipated, providing a peace of mind knowing that customers will still be able to access and use the website even if failure occurs.

### CloudWatch

In addition, CloudWatch can also be implemented as a monitoring system for the website. Through CloudWatch, real time network monitoring is enabled allowing data to be collected from the EC2 instances and load balancers to allow Sophia and Nikhil to easily gather insights on their website's performance and how well customer demand is handled. Custom alerts can also be set, where messages are sent through Amazon SNS to notify Sophia and Nikhil of issues that are occurring, offering a proactive way to address issues quickly. Traffic patterns can also be identified through the CloudWatch Internet Monitor which is able to identify trends in customer issues based on geographic locations to provide recommendations on how latency can be reduced by switching to regions where customer traffic is higher (AWS n.d.).

By monitoring instance performance in almost real time and sending alerts to Sophia and Nikhil, they are able to promptly respond to deviations and make frequent informed decisions to maximise efficiency and boost operational excellence.

### Security Through Bastion Host and WAF

For extra security, Bastion Host and Web Application Firewall (WAF) can also be implemented. Both services can be added to CloudFront Distributions to enhance security through backend resource access control and managing malicious requests from the frontend. Through this firewall service designed to protect websites from malicious attacks, AWS managed rule sets can be used to help protect the cafes website and resources, prevent data breaches and website downtime due to attacks. For future reference, Bastion Host can also be implemented to minimise the exposure of resources by providing secure access to private subnets through the public subnet of the Amazon VPC (Malaval 2016).

Through Bastion Host, strong identity foundation and data protection is enabled through tight access controls, allowing only authenticated users to connect to private instances through inbound security rules set up through SSH keys and IP address rules, allowing only users with the specified IP address to access the bastion host. WAF and Bastion Host also add to the security layer with WAF filtering traffic requests and the latter adding a security layer between public and private subnets to ensure data confidentiality and integrity.

### **Implementation run-sheet**

#### **Task 1: Updating a network to work across multiple Availability Zones**

- In the search box next to Services, search for and select **VPC**.
- Create a NAT gateway in the Public Subnet in the second Availability Zone.
  - o In the left navigation pane, choose **NAT gateways** and **Create NAT gateways**.
  - o Name: lab- natgateway
  - o Subnet: **Public Subnet 2**
  - o Keep the default connectivity type **Public** selected
  - o Click on **Allocate Elastic IP**
    - This NAT gateway allows EC2 instances in private subnets to access the internet and must be done before instances are launched to EC2 in the second availability zone.
- Send internet-bound traffic to the NAT gateway that was just created.
  - o In the left navigation pane, choose **Route tables**.
- Select **Private Route Table 2**.
- Choose the **Routes** tab under the screen and **Edit routes**.
- Click **Add route** and configure the following:
  - o Destination: 0.0.0.0/0
  - o Target: NAT Gateway
  - o Search: nat-xxxxxxxxxx (lab-natgateway)

Note: The Bastion Host would be implemented after this step.

#### **Task 2: Creating a launch template**

- In the search box next to Services, search for and select **EC2**.
- In the left navigation pane find the Instances section and click **Launch Templates**.
- Create a launch template. (refer to challenge lab for further details)

#### **Task 3: Creating an Auto Scaling group**

- In the left navigation pane of the EC2 console, click **Auto Scaling Groups**.
- Create autoscaling group. (refer to challenge lab for further details)
- After creating an autoscaling group, in the left navigation pane of the EC2 console, click **Instances** and the refresh button.
- Check if there are two instances named webserver.

#### **Task 4: Creating a load balancer**

- In the left navigation pane find the Load Balancing section and click **Load Balancers**.
- Create a load balancer.
  - o **Security groups**: Create a new security group.
    - Name: lab-securitygroup
    - Description: **Allows only HTTP traffic**

- VPC: **Lab VPC**
- Inbound rules
  - Type: **HTTP**
  - Source: **Anywhere-IPv4**
- o Remove the default security group, and select **lab-securitygroup**.
- o Under the Listeners and routing section, click **Create target group**.
  - Target group name: lab-targetgroup
- o Refresh the default action, and select lab-targetgroup
- Refresh the load balancer until the status turns to active.
- Modify the Auto Scaling group by adding the new load balancer.
  - o In the left navigation pane of the EC2 console, click **Auto Scaling Groups**.
  - o Select the name of the group that was just created.
  - o Scroll down to the **Load Balancing** section and select **Edit**.
    - Select **Application, Network, or Gateway Load Balancer target groups**.
    - Select **lab-targetgroup** and **Update**.

Note: In the future availability zones should be switched to the eu-west-2 London region to reduce latency, as most of the customers accessing the website would be located locally in London.

# Architecture Diagram

The first iteration of the AWS architecture diagram [Diagram 1] shows the most basic and simplest way of creating a scalable and highly available website for the Café to fulfil the requirements of the lab. The current security practices of the level 1 Architecture diagram only use basic security groups and NAT gateways for private instances. The website would benefit from a more robust security configurations to protect it against malicious users and traffic. Additionally, the current database instance has limited storage and does not accommodate for the Café’s data growth or the potential for high read and write traffic that might result from the upcoming exposure on the TV show. The current setup lacks scalability, which could make it a single point of failure. If the database instance fails or if demand exceeds its capacity, the Cafe’s website could experience significant downtime or delays, impacting customer experience and the business’s reputation.

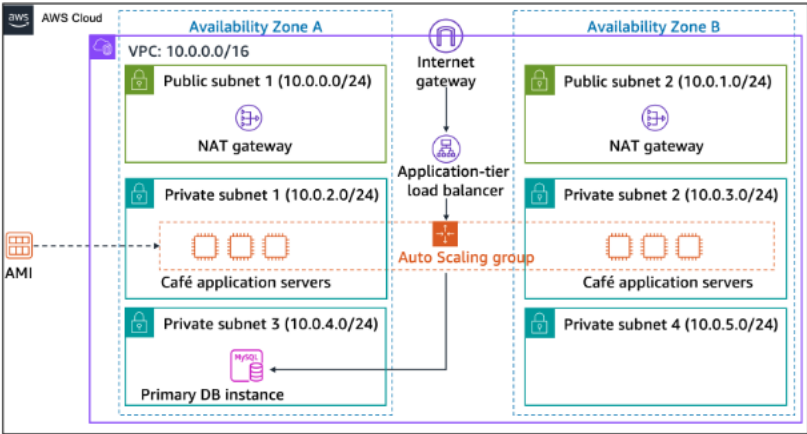


Diagram 1 – First Iteration of AWS architecture diagram

The improved AWS Architecture diagram was implemented to enhance the scalability, security and resilience of the Café’s website, addressing the limitations of the level 1 design. With an expected increase in traffic due to a TV show feature, the architecture required needs to be more robust and can handle large volumes of users. This new wave of exposure also raises the concerns of potential threats to the website. By adding AWS WAF, Amazon CloudFront, Bastion Host and Amazon RDS as a backup database, the architecture improves both performance and security. The reasons for adding these features were mentioned in the business decisions.

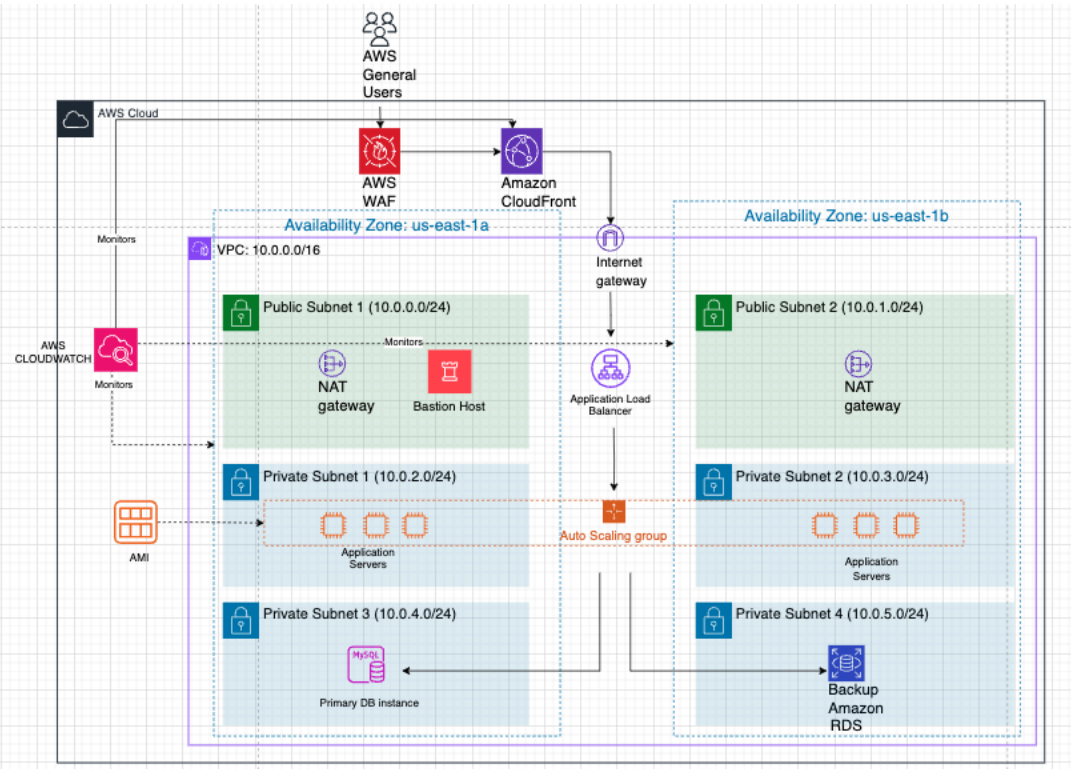


Diagram 2 – New and improved AWS architecture diagram

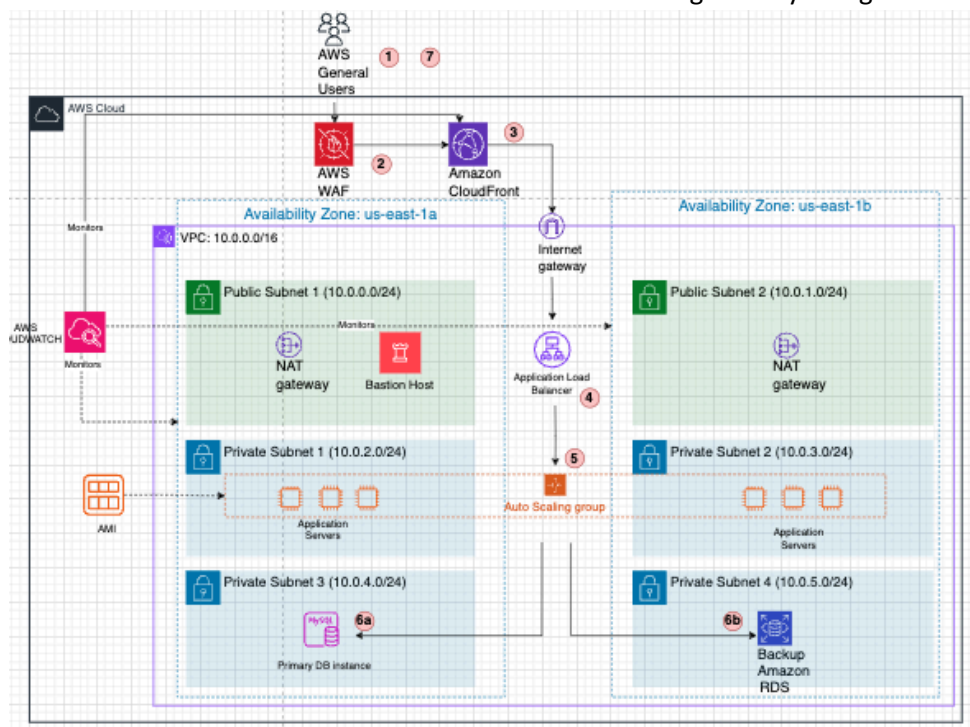
AWS WAF acts as the first line of defense against malicious web traffic (AWS n.d) It is deployed in front of the Application Load Balancer (ALB) and filters incoming requests before they can reach the web servers. AWS WAD is linked with Amazon CloudFront and then the ALB. Since CloudFront is set up to cache static content, AWS WAF inspects incoming requests at the edge before they reach the ALB. This prevents malicious requests from reaching the Cafe’s website altogether. The next level is the ALB, directing legitimate traffic to the appropriate Café application servers. Another linkage is with Bastion Host

with the broader security access to add another level of entry point security for admins of the website.

The linkages between the autoscaling group and Amazon RDS (the new backup database) works together when web traffic increases, the auto scaling group can automatically add more EC2 instances to handle the additional load. These new instances rely on the Amazon RDS database for consistent access to the application's data. The backup database acts as a safety net in case the primary database becomes unavailable or compromised. This improves resilience of the website by ensuring that the system has a backup database ready to be used without disrupting the application's performance or availability.

Another link is CloudWatch and the services contained used for the website. CloudWatch monitors metrics and tracks the performance of AWS resources. For example, when CloudWatch and Auto Scaling groups link together it monitors metrics like CPU utilization, memory usage, or request count from the EC2 instances, using this information to trigger Auto Scaling actions, like launching or terminating instances to maintain the desired performance and availability of the website.

For further iterations of the café website and AWS architecture diagram, the owners may want to consider using Amazon Route 53 to create a custom website domain name making it easily recognizable to customers looking for their website.



To show the flow at which users access the website, and the services involved in the process:

1. When a customer first accesses the website, their request is routed through AWS WAF to check if their network traffic is flagged for malicious activities.
2. After passing this check the request is redirected to Amazon CloudFront, which acts as a Content Delivery Network (AWS n.d)
3. Once the static content is loaded, the customer submits an order. This request is sent to the ALB
4. The ALB forwards the customer's request to one of the EC2 instances within the ALB and then the ALB routes this request to the appropriate EC2 instances.
5. As the customer's order is processed, the auto scaling group assigns the request to a application server.
6. EC2 servers then interacts with the Amazon database to store or retrieve data related to the order
  - a. This is the primary database where all requests will be stored until it is full
  - b. Requests will be stored here is the primary database becomes compromised
7. The final request message will then be sent back to the customers

Note: During this entire process CloudWatch will be actively monitoring the performance of all AWS services and tools.

# Cost Estimate

Assumption: Because it is still a fairly small café we are Assuming there will be low to moderate traffic on the website every day so there will be an average of 3gb of data being transferred from the website everyday – rough estimate (3.4kb of data transfer bandwidth every time you open the website)

0 B / 3.4 kB transferred

| Service Name                       |  | Upfront cost | Monthly cost |
|------------------------------------|--|--------------|--------------|
| Amazon Virtual Private Cloud (VPC) |  | 0.00 USD     | 36.90 USD    |
| Amazon EC2                         |  | 0.00 USD     | 15.04 USD    |
| Amazon RDS for Db2                 |  | 0.00 USD     | 23.00 USD    |
| AWS Web Application Firewall (WAF) |  | 0.00 USD     | 9.60 USD     |
| Amazon CloudWatch                  |  | 0.00 USD     | 25.69 USD    |
| Elastic Load Balancing             |  | 0.00 USD     | 17.17 USD    |
| AWS Data Transfer                  |  | 0.00 USD     | 8.10 USD     |

Monthly cost

135.50 USD

Total 12 months cost

1,626.00 USD

Includes upfront cost

## AWS Data Transfer (\$8.10/month assumed)

Our architecture is designed to decrease outbound data charges, assuming that there is 3GB of daily data transfer for the Cafe, internet inbound data transfers have been set to 100GB, whereas outbound data transfers have been set to 90GB. Reducing the data transfer allocation could induce risk, particularly during peak times, when the Cafe is expecting tens of thousands of application users. Sufficient data transfer allowance can prevent unexpected, incurred costs, particularly when data transfer costs are unavoidable for an internet application expecting a large number of users during peak times.

## Amazon Virtual Private Cloud (VPC) (\$36.90/month)

VPC allows us to establish a secure network environment through configuring a NAT Gateway, which allows for private instances to gain access to the internet discreetly. The cost is negligible, considering the security it provides in allowing private subnets to access the internet securely.

Instead of using the NAT Gateway feature, individual public IPS could have been configured for all instances, however this lack of network segmentation could pose a significant security risk, as well as incur a more laborious setup for the application. Thus, by configuring a NAT Gateway, the Cafe’s business goal of creating a scalable, and highly secure network architecture is upheld.

## Amazon EC2 (\$15.04/month)

Notable configurations include shared instances, t2.micro instance type, and a 3-year no upfront pricing strategy. Choosing shared instances decreases infrastructure and hardware costs, suitable for the Cafe’s desire to be scalable, yet likely budget conscious. The Linux operating system is also configured for its compatibility with many applications. The selection of the t2.micro instance allows for a balance between a moderate to high performance, and cost-effective set-up. Monitoring is disabled, as it will be touched upon in the usage of AWS CloudWatch, which will perform similarly.

## Elastic Load Balancing (\$17.17/month)

Elastic Load Balancing is a significant feature to allow for traffic distribution amongst multiple instances. It allows for increased traffic loads during peak times to be handled effectively, ensuring scalability and availability of the application is met.

While a cheaper option is to host a single server, this option provides a minimally scalable and available approach, potentially leading to downtimes under high traffic load, therefore load balancing is implemented to handle fluctuating traffic.

Amazon RDS for Db2 (\$23.00/month)

Multi-AZ is configured to provide high availability, and to improve system resilience against failures in single Availability zones. Instance type db.t2.micro provides sufficient memory while also being cost-effective, deterring any unnecessary costs. While using the Single-AZ deployment is lower in cost, it compromises high availability. Thus, these configurations offer a balance between high performance and minimal cost.

Amazon CloudWatch (\$25.69/month)

The chosen configuration includes 5 metrics, 10 GetMetricData requests, a standard log ingestion of 3GB, and 20 API requests. CloudWatch allows for continuous real-time monitoring and alerting on the applications performance and health status, which is crucial for highly available set-up. Key metrics and basic log ingestion are required for an internet-facing application expecting moderate to high traffic. Therefore, while costs can be cut by reducing the metrics and logs available, it would severely limit access to viewing the system’s health and hinder continuous monitoring, which could be detrimental for the Cafe’s applications availability.

AWS Web Application Firewall (WAF) (\$9.60/month)

WAF is essential for mitigating potential security threats, as it provides an extra layer of security and protection through its monitoring of requests to the public-facing application. The use of WAF could be deterred to lower costs, however, for a customer-facing application, the minimal cost per month is justified to prevent web exploits and to better control the application traffic.

Furthermore, implementation of Bastion Host for SSH is beneficial for the cafe, enhancing isolation of instances. It acts as a secure gateway for administrators to gain controlled access to private instances within the cloud. By utilising Bastion, SSH access is limited only specifically to the bastion instance, adhering to security principles of least privilege, and ensuring that only access deemed essential is permitted.

t2.micro

AWS Free Tier

If you qualify for the [AWS Free Tier](#), Amazon EC2 charges for Micro instances are free for up to 750 hours a month.

t2.small

t2.medium

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